

Alignment Dynamics in LLM Fine-Tuning

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Abstract

Although Large Language Models (LLMs) achieve strong alignment through supervised fine-tuning and reinforcement learning from human feedback, the alignment is often fragile under subsequent fine-tuning. Existing explanations either attribute alignment fragility to gradient geometry or characterize it as a distributional shift in model outputs, yet few provide a unified account that bridges parameter-space learning dynamics with function-space alignment behavior during fine-tuning. In this work, we introduce a tractable alignment score and derive its closed-form update during fine-tuning, yielding a unified framework for alignment dynamics. Our analysis decomposes alignment updates into two competing components: a **Rebound Force**, governed jointly by the current alignment state and the narrowness of model distribution, and a **Driving Force**, determined by how the training distribution aligns with outcome-conditioned posteriors over aligned and non-aligned completions. This decomposition explains why prior alignment can be reversed by later fine-tuning and why narrower posterior structure strengthens such reversal. Moreover, our framework predicts a **Rehearsal Priming Effect**: prior alignment leaves a latent posterior imprint that amplifies the effective Driving Force upon re-exposure, leading to faster re-alignment. We validate these predictions across safety alignment, emergent misalignment, and sentiment settings, demonstrating consistent alignment reversal and accelerated re-alignment under re-exposure. In addition, controlled experiments in safety alignment confirm the predicted dependence of rebound strength on posterior narrowness. Together, these results provide a unified dynamical perspective on how alignment is disrupted and reactivated during LLM fine-tuning.

1 Introduction

Large Language Models (LLMs) have achieved remarkable success in aligning their behavior with human values through a combination of supervised fine-tuning, preference optimization, and post-training techniques [1, 2, 3, 4, 5, 6, 7]. Despite this progress, alignment remains persistently fragile, posing a major obstacle to reliable deployment. Recent studies show that safety guardrails can be easily compromised by fine-tuning on small amounts of harmful data, and even on benign data [8, 9, 10]. This pervasive vulnerability underscores the urgent necessity for a unified theoretical framework that moves beyond empirical observations to provide a rigorous, quantitative characterization of how alignment evolves, degrades, and recovers during the fine-tuning process.

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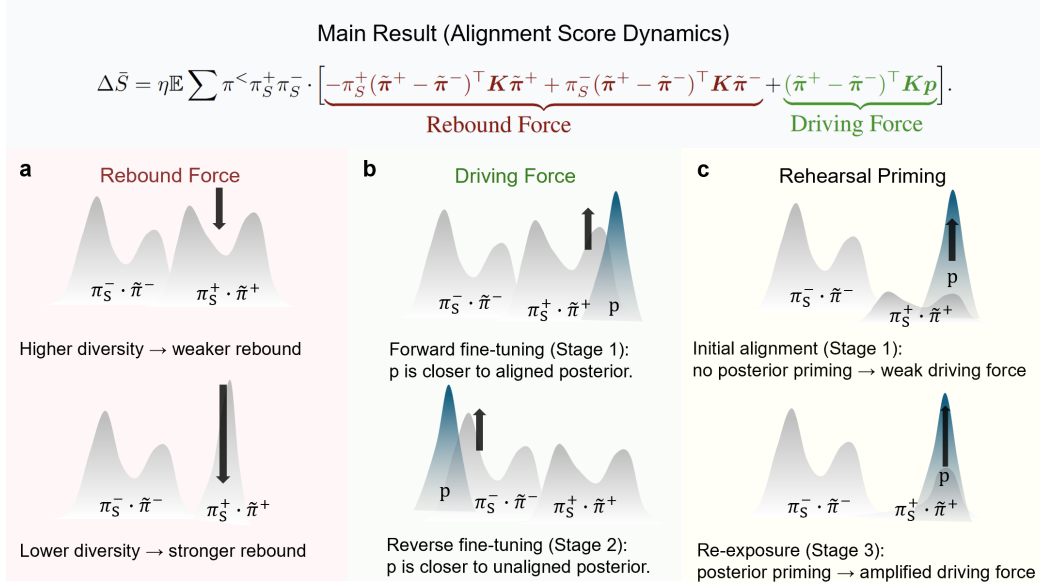


Figure 1: Illustration of alignment dynamics in LLM fine-tuning. (a) **Rebound Force**: The strength of the intrinsic rebound effect is determined by the narrowness of the posterior distribution, with lower diversity leading to stronger rebound force. (b) **Driving Force**: Alignment evolution is governed by the relative alignment of the fine-tuning distribution p with the outcome-conditioned posteriors $\tilde{\pi}^+$ and $\tilde{\pi}^-$ over aligned and non-aligned completions. (c) **Rehearsal Priming Effect**: Prior fine-tuning reshapes the posterior structure, which amplifies the effective driving force upon re-exposure and leads to faster re-alignment.

Existing explanations for alignment fragility can be categorized into gradient-based and distribution-based perspectives. Gradient-based analyses [11, 12, 13] base on first-order geometry to decompose fine-tuning gradients into safety-relevant subspaces, attributing degradation to insufficient projection onto safety-aligned directions. While geometrically intuitive, these analyses do not explicitly map parameter-level updates to functional changes in output probabilities, and thus leave unclear how data distributions shape the resulting dynamics. On the other hand, distribution-based studies characterize alignment as a shallow modification of the model’s output distribution. This perspective identifies disruption mechanisms such as compression-driven reversion toward the pre-training distribution [14], distributional reweighting [15, 16], and asymmetric forgetting of safety-relevant behaviors [17]. However, this line of work remains largely static: it describes the resulting distributional shift, but does not explain the optimization dynamics through which misalignment emerges during fine-tuning.

Taken together, these perspectives underscore a central challenge of how to bridge parameter-space learning dynamics with function-space alignment behavior. While prior work has begun to characterize one-step probability dynamics [18], it remains unclear how to define alignment in a sufficiently general yet tractable form, and how such a term evolves under fine-tuning. We address this challenge by introducing a precise alignment score S and deriving its closed-form evolution under learning dynamics. The resulting decomposition reveals a non-trivial relationship between alignment change and outcome-conditioned posterior distributions over aligned and non-aligned completions. In the case of supervised fine-tuning, this decomposition further exposes two competing forces, a **Rebound Force** and a **Driving Force**, as shown in Figure 1.

The **Rebound Force** ($\mathcal{F}_{\text{rebound}}$) is an intrinsic self-interaction term of the model’s conditional output distribution. It is modulated by both the current alignment state and the sharpness of the posterior distribution. This structure implies a bidirectional instability: both strongly aligned and strongly misaligned states become more reversible under subsequent perturbations (Section 3.2). Moreover, our framework predicts that narrower posterior distributions, which are induced by lower-diversity fine-tuning data, amplify the rebound effect. We empirically establish this relationship across controlled fine-tuning settings with varying data diversity (Section 3.3).

The **Driving Force** ($\mathcal{F}_{\text{drive}}$) captures the external learning signal induced by fine-tuning. It is determined by how the training distribution $\mathbf{p}$ differentially aligns with the two outcome-conditioned posteriors corresponding to aligned and non-aligned completions. As a result, even a small change in the model’s surface probabilities can yield a large change in alignment speed if it substantially reshapes the posterior structure. In this sense, prior alignment may leave only a modest footprint in observed probabilities while still strongly accelerating subsequent re-alignment.

This leads to a new phenomenon, which we term the **Rehearsal Priming Effect**: prior alignment leaves a latent posterior imprint, so that re-exposure to alignment data triggers substantially faster re-alignment than during the initial alignment stage, even after subsequent fine-tuning has weakened the observed safety behavior. We empirically demonstrate that this effect arises consistently across diverse settings, including safety alignment, emergent misalignment, and sentiment, indicating that it reflects a general property of alignment dynamics rather than a task-specific artifact (Section 4).

2 A Unified Formula for Alignment Dynamics

We now derive a unified formula for alignment dynamics under fine-tuning. We begin by formalizing the alignment score, then decompose its evolution into token-level contributions, and finally characterize the resulting dynamics as the interplay between driving and rebound forces. We provide a proof sketch in this section to highlight the core ideas, with details deferred to Appendix A.

2.1 Preliminaries

Notations. Let $\mathcal{V}$ denote a finite vocabulary of size V . We consider a sequence-to-sequence task where a prompt $\mathbf{x} := [x_1, \dots, x_{L_x}] \in \mathcal{V}^{L_x}$ is drawn from a data distribution $p(\mathbf{x})$. For each prompt $\mathbf{x}$, a language model parameterized by $\theta \in \mathbb{R}^d$ generates a completion $\mathbf{y} := [y_1, \dots, y_{L_y}] \in \mathcal{V}^{L_y}$. We denote the space of all possible prompts and completions as $\mathcal{X} := \mathcal{V}^{L_x}$ and $\mathcal{Y} := \mathcal{V}^{L_y}$, respectively.

Autoregressive Policy. The model defines a conditional distribution $\pi_\theta(\mathbf{y} \mid \mathbf{x})$, which, due to the autoregressive nature of the task, factorizes into a product of token-level conditional probabilities:

$$\pi_\theta(\mathbf{y} \mid \mathbf{x}) = \prod_{l=1}^{L_y} \pi_\theta(y_l \mid \mathbf{x}, \mathbf{y}_{<l}), \quad (1)$$

where $\mathbf{y}_{<l} := [y_1, \dots, y_{l-1}]$ represents the prefix sequence generated prior to step l , with $\mathbf{y}_{<1}$ being an empty sequence.

The Aligned Set. To formalize the notion of alignment (e.g., safety, helpfulness, or style), we define an aligned subset $\mathcal{A} \subseteq \mathcal{X} \times \mathcal{Y}$. A prompt-completion pair $(\mathbf{x}, \mathbf{y})$ is considered aligned if and only if $(\mathbf{x}, \mathbf{y}) \in \mathcal{A}$. In practical scenarios, the membership of $\mathcal{A}$ is typically determined by human preference or a calibrated judge model.

Definition 2.1 (Alignment Score). Given a policy π_θ and an aligned subset $\mathcal{A}$, the alignment score $S(\theta)$ is defined as the expected probability of the model generating a completion that satisfies the alignment criteria for a prompt drawn from $p(\mathbf{x})$:

$$S(\theta) := \mathbb{E}_{\mathbf{x} \sim p(\mathbf{x})} \left[\sum_{\mathbf{y} \in \mathcal{Y}} \mathbb{I}_{(\mathbf{x}, \mathbf{y}) \in \mathcal{A}} \cdot \pi_\theta(\mathbf{y} \mid \mathbf{x}) \right], \quad (2)$$

where $\mathbb{I}$ is the indicator function.

2.2 Token-level Decomposition of Alignment Dynamics

While the alignment score $S(\theta)$ measures the global adherence of the model to the aligned set $\mathcal{A}$, the optimization of LLMs occurs through local updates to token-level conditional distributions. To understand the mechanism of alignment, it is crucial to determine how the shift in a single token’s probability contributes to the overall evolution of $S(\theta)$. This necessitates a decomposition that assigns “credit” or “responsibility” for alignment progress to specific positions within a sequence.

To facilitate this decomposition, we introduce two auxiliary variables that characterize the state of generation at step l . For a prompt-completion pair $(\mathbf{x}, \mathbf{y})$, we define: